

Project Write-up:

**BUREAUCRACY, BUDGETS, AND THE GENDER GAP:
A GLOBAL AND LOCAL PERSPECTIVE****Group 17:** Phan Thi Hien Chi - Nguyen Mau Hoang Hiep - Pham Minh Hieu

1 Introduction

This project uses the *Worldwide Bureaucracy Indicators (WWBI)* dataset [2], developed by the World Bank's Bureaucracy Lab and curated by the TidyTuesday project [3]. The WWBI is a large cross-national dataset on public sector employment and wages, compiled from administrative records and household surveys. It includes 192 indicators across 202 countries and territories between 2000 and 2020. By encompassing detailed information on public-sector employment characteristics, wages and compensations, and overall wage bills, the WWBI provides researchers with a robust foundation for comparative analyses of government workforce structures, compensation practices, and their evolution over time. Although the WWBI offers comprehensive coverage, it does face certain limitations, including missing values for some countries and years, potential inconsistencies arising from varying data collection methodologies, and limited information on informal employment. Nevertheless, it remains the most extensive resource currently available for conducting meaningful comparative analyses of government employment practices worldwide. Our motivation for selecting this dataset stems from this exceptional breadth and depth in documenting public sector employment patterns, making it uniquely suited for examining critical questions about governance, economic development, and gender equity in government workforces.

In this project, we aim to explore two research questions: (1) *How has the proportion of GDP allocated to public wages changed over time across different world regions and income groups?*, and (2) *How has the gap in female representation between high-skill and low-skill occupations¹ evolved over time in Vietnam, and how does it differ between the public and private sectors?* These questions address fundamental aspects of public governance efficiency, economic resource allocation, and gender equity in workforce participation — issues with significant implications for sustainable development, economic stability, and social cohesion. They are especially relevant in both global and Vietnamese contexts, particularly for young people like us who are preparing to enter the workforce. To address these questions, we analyze key variables such as: **year** (temporal dimension spanning 2000–2020); **country_code** (ISO 3166-1 country identifiers); **indicator_code** (including `BI.WAG.TOTL.GD.ZS` for wage bill as a percentage of GDP, and various employment indicators categorized by sector, occupation level, and gender); **value** (numeric measurements); **region** (geographical classification); and **income_group** (economic development classification). Further description of the variables and analysis for each question will be provided in the following sections.

The code for our project is available at <https://github.com/chipphan1110/wwbi-visualization.git>.

2 Question 1: How has the proportion of GDP allocated to public wages changed over time across different world regions and income groups?

2.1 Introduction

In this analysis, we aim to explore how the proportion of GDP allocated to public wages has changed over time across different world regions and income groups. Understanding public sector wage spending is important because it reflects a government's fiscal priorities, its commitment to public services, and its broader economic health. Given the ongoing debates about public sector efficiency and fiscal responsibility worldwide — and particularly in Vietnam recently — examining these trends can reveal valuable insights into how different economies manage their public wage bills under various economic and political pressures.

To address the question, we prepare the dataset by merging `wwbi_data.csv`, `wwbi_series.csv`, and `wwbi_country.csv`. We then filter the data to retain only entries for the indicator: `BI.WAG.TOTL.GD.ZS` — Public sector wage bill as a percentage of GDP and use metadata from `wwbi_country.csv` to map each country to its corresponding **region** and **income_group** classification. From that, we focus on analyzing these variables: **year**, **country_code**, **income_group**, **region**, and **value** for the chosen indicator.

¹High-skill and low-skill occupations are classified according to ISCO-08 skill levels, following the International Labour Organization's definitions. See <https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/>.

2.2 Approach

To address this question, we decided to create a line plot with facets by region with color mapping by income group. Each line represents an individual country's trend over time, while smoothed average trends by income group are shown in bold, with shaded confidence intervals. This plot will show the trend of public wage as a percentage of GDP over time for different world regions. Faceting by region allows for clear and focused comparisons across different parts of the world, while color mapping enhances the visibility of differences between income groups within each region. Line plots are particularly well-suited for this task because they effectively display temporal trends and highlight variations both within and across groups. The addition of smoothed trend lines and confidence intervals helps emphasize general patterns while accounting for year-to-year fluctuations and data uncertainty. The specific and grouping in the visualization are:

- Income group: high income, upper middle income, lower middle income, low income.
- Regions: Europe & Central Asia, Middle East & North Africa, East Asia & Pacific, South Asia, Sub-Saharan Africa, Latin America & Caribbean, North America.

Before visualization, we apply a logarithmic transformation to the wage bill values. This helps to reduce the skewness in the data, allowing better visualization and comparison across countries with very different GDP structures. It also makes relative differences clearer, particularly for countries with smaller wage bill percentages.

2.3 Analysis

The below block is our main code for the grouped bar chart with line overlay. For more detailed visualization and formatting, please review [our code](#). The result visualization is shown in [Figure 1](#).

```
# Plot individual country lines
for country, country_data in region_data.groupby('country_code'):
    income_group = country_data['income_group'].iloc[0]
    country_data = country_data.sort_values('year')
    ax.plot(
        country_data['year'],
        country_data['value'],
        color=palette[income_group],
        linewidth=0.25,
        alpha=0.7
    )

# Plot smooth trend lines
sns.lineplot(
    data=region_data,
    x='year',
    y='value',
    hue='income_group',
    palette=palette,
    linewidth=2.0,
    alpha=0.6,
    ax=ax,
    errorbar=('ci', 80),
    err_style='band',
    err_kws={
        'alpha': 0.15,
        'linewidth': 0,
    },
    estimator=lambda x: np.mean(x),
    n_boot=1000,
    legend=False if i > 0 else True
)
```

2.4 Discussion

Our visualization reveals clear patterns in how public wage spending as a percentage of GDP varies across different regions and income groups from 2000 to 2020. In general, high-income countries consistently maintain relatively stable wage bills, typically within the range of 8%–15% of GDP. In contrast, low-income countries tend to allocate a smaller proportion of their GDP to public wages, often falling below 8%. These results align with expectations, as higher-income countries tend to have more established public sectors and the fiscal capacity to sustain larger wage bills relative to GDP.

Wage Bill as Percentage of GDP by Region and Income Group

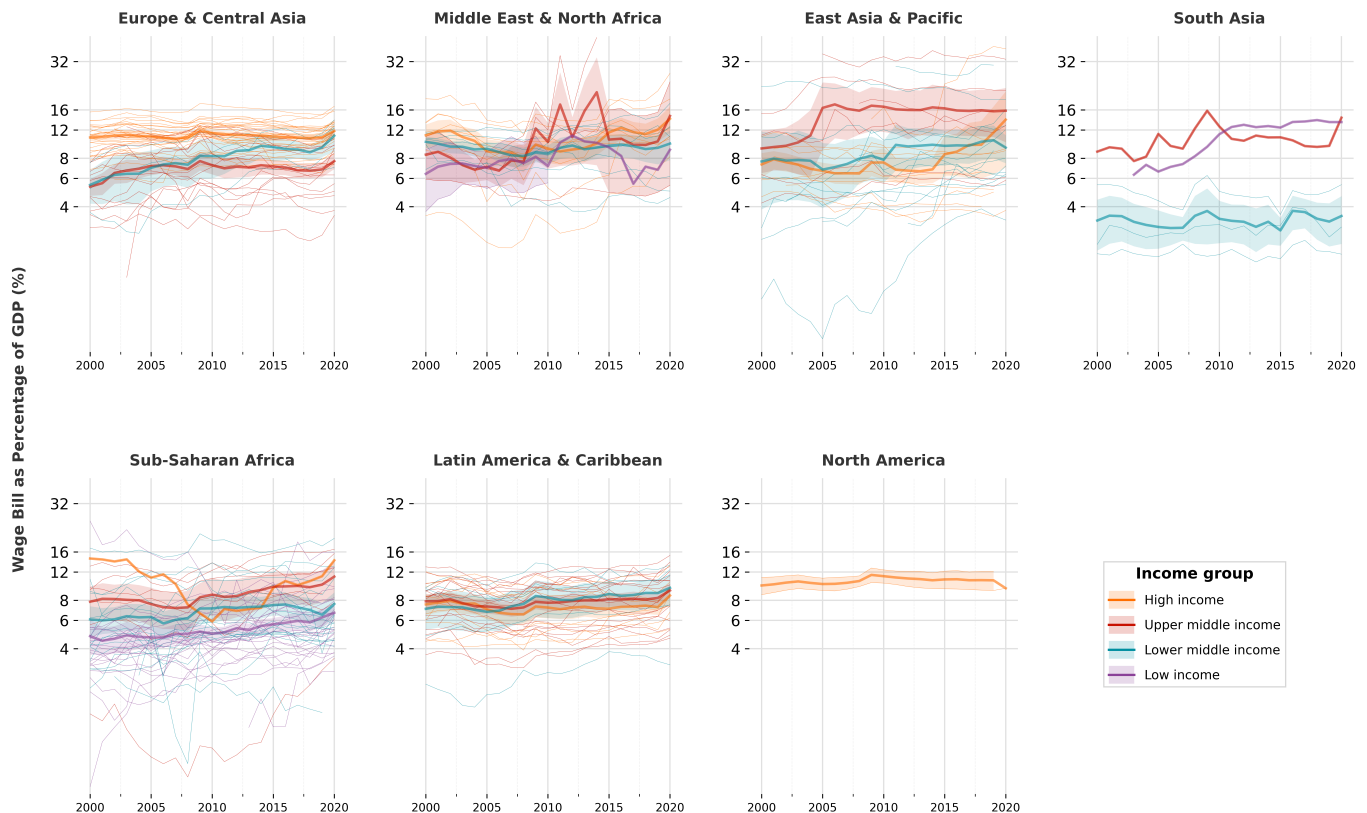


Figure 1: Trends in the public sector wage bill as a percentage of GDP across different world regions and income groups from 2000 to 2020. Light lines represent country-level data, while bold lines show smoothed trends for each income group with a 95% confidence interval (shaded area) estimated using Seaborn's smoothing function.

We also observe interesting regional variations. For example, Latin America and the Caribbean shows remarkably stable trends over the two decades, with little fluctuation in wage bill levels. In contrast, East Asia and the Pacific exhibits more dramatic changes, including periods of rapid increase, likely reflecting broader political-economic shifts and evolving government structures during this time. Similarly, Middle East and North Africa displays noticeable fluctuations, particularly around 2010–2015, possibly influenced by regional instability, the Arab Spring, and fiscal adjustments. These patterns suggest that political events, economic reforms, or external shocks can significantly impact public sector wage spending.

Additionally, the plots highlight that while high-income groups maintain stability, lower-income groups show more volatility and wider confidence intervals, indicating greater variability across countries. This variability might reflect challenges faced by developing nations, including fiscal constraints, changing governance priorities, and efforts to expand or reform public services. Overall, the analysis emphasizes the importance of considering both economic development level and regional context when evaluating trends in public wage spending. Policymakers, especially in developing regions, could benefit from this understanding by designing public sector policies that are more resilient to political and economic shocks, while ensuring sustainable management of public wage bills.

3 Question 2: How has the gap in female representation between high-skill level and low-skill level occupations evolved over time in Vietnam, and how does it differ between the public and private sectors?

3.1 Introduction

In this analysis, we explore how the gap in female representation between high-level and low-level occupations has evolved over time in our country, Vietnam, and how it differs between the public and private sectors. Gender equity in employment is a critical issue globally and in Vietnam, where promoting women's participation in leadership roles is increasingly recognized as essential for sustainable economic development [1]. As university students preparing to enter

the workforce, we are eager to understand the structural barriers that could influence our future career opportunities. Through this exploration, we aim to gain insights and awareness about these challenges and recognize our responsibility in advocating for a more equitable and inclusive society

Similar to question 1, to address this question, we use the `wwbi_data.csv` file, which contains public-sector employment indicators, and supplement it with `wwbi_series.csv` and `wwbi_country.csv` to map indicator descriptions and country information. We filter the data to retain only entries for Vietnam (`'country_code' = 'VNM'`) and select specific indicators measuring the share of females among paid employees across different occupation categories. The selected variables are:

- `year`
- `value`
- `indicator codes`
 - Public sector: `BI.PWK.PUBS.SN.FE.ZS`, `BI.PWK.PUBS.PN.FE.ZS`, `BI.PWK.PUBS.TN.FE.ZS`, `BI.PWK.PUBS.CK.FE.ZS`, `BI.PWK.PUBS.EO.FE.ZS`
 - Private sector: `BI.PWK.PRVS.SN.FE.ZS`, `BI.PWK.PRVS.PN.FE.ZS`, `BI.PWK.PRVS.TN.FE.ZS`, `BI.PWK.PRVS.CK.FE.ZS`, `BI.PWK.PRVS.EO.FE.ZS`

with `BI.PWK.<sector>.<occupation>.FE.ZS` indicate the proportion of females in a specific occupation group relative to all paid employees in that group, either in the public (PUBS) or private sector (PRVS). We further group occupations into high-skill level and low-skill level categories following the ISCO-08 classification with specific code below:

- High-Level Roles: Senior Managers (SN), Professionals (PN), Technicians (TN)
- Low-Level Roles: Clerks (CK), Elementary Workers (EO)

3.2 Approach

Regarding data processing after filtering, we aggregated the data by calculating the average female share in each broad role group (high-level and low-level) for each year and sector. We computed the gender gap as the difference between the average female share in high-level and low-level occupations by: $\text{Gap} = \% \text{High-Level} - \% \text{Low-Level}$. This gap provides a direct measure of gender inequality across occupational skill levels over time. Then, we designed two visualizations to explore the trends:

- **Grouped Bar Chart with Line Overlay (Faceted by Sector):** This plot shows the percentage of females in high-skill and low-skill occupations over time, with a line overlay representing the gap between high- and low-skill female representation. The grouped bars allow clear comparison between high and low categories at each time point, while the line highlights trends in the gap between representation in high and low-level occupation. The shaded area gives a very straightforward intuition - the larger the area, the larger the gap. Faceting by sector (public vs. private) enables direct comparison across different labor market segments. The combination of bar plot and line overlay (with shaded area under the line) provides a comprehensive view of proportions and inequalities over time.
- **Lollipop Plot (Faceted by Sector and Colored by Occupation):** To examine more detailed occupation types, we decided to use a lollipop plot showing female representation percentages across managers, professionals, technicians, clerical workers, and elementary workers. Lollipop plots offer a clean and uncluttered way to compare values across multiple categories. Color mapping by occupation type makes it easy to distinguish different roles, while faceting by sector helps identify differences between public and private employment structures.

3.3 Analysis

The below block is our main code for the grouped bar chart with line overlay. For more detailed visualization and formatting, please review [our code](#). The result visualization is shown in [Figure 2](#).

```
# Plot for each sector
for i, sector in enumerate(['Public', 'Private']):
    sector_data = level_data[level_data['sector'] == sector]

    # Get all years
    years = sorted(sector_data['year'].unique())

    # Plot High and Low level bars
    for j, level in enumerate(['High', 'Low']):
```

```

level_subset = sector_data[sector_data['level'] == level]

# Create a mapping from year to value
year_to_value = {row['year']: row['value'] for _, row in level_subset.iterrows()}

# Get values for all years (use 0 if data is missing)
values = [year_to_value.get(year, 0) for year in years]

# Plot the bars
bar_positions = x_positions + (j - 0.5) * bar_width
bars = axes[i].bar(bar_positions, values, width=bar_width,
                  label=f'{level}-Level', color=colors[level], alpha=0.85)
for bar in bars:
    # Add percentage labels on top of the bars
    height = bar.get_height()
    axes[i].annotate(f'{height:.1f}%',
                    xy=(bar.get_x() + bar.get_width() / 2, height),
                    xytext=(0, 3), textcoords="offset points",
                    ha='center', va='bottom', fontsize=10, color='#333', fontweight='bold')

# Calculate and plot the gap between high and low
high_values = sector_data[sector_data['level'] == 'High'].set_index('year')['value']
low_values = sector_data[sector_data['level'] == 'Low'].set_index('year')['value']

# Create a complete index of all years
all_years = pd.Index(years)
high_values = high_values.reindex(all_years).fillna(0)
low_values = low_values.reindex(all_years).fillna(0)

# Calculate the gap
gap_values = high_values - low_values

# Fill area under gap line
gradient_fill(axes[i], x_positions, gap_values.values, cmap=gradient_cmap, alpha=0.3)

# Plot the gap as a line
axes[i].plot(x_positions, gap_values, 'k--', marker='o', color='Orange', label='Gap (High - Low)')

```

And this below block is our main code for the lollipop chart. For more detailed visualization and formatting, please review [our code](#). The result visualization is shown in [Figure 3](#).

```

# Plot for each sector
for i, sector in enumerate(['Public', 'Private']):
    sector_data = grouped[grouped['sector'] == sector]
    years = sorted(sector_data['year'].unique())
    # Plot for each occupation
    for j, occupation in enumerate(occupations):
        occ_data = sector_data[sector_data['occupation'] == occupation]
        year_to_value = {row['year']: row['value'] for _, row in occ_data.iterrows()}
        values = [year_to_value.get(year, 0) for year in years]
        pos = x_positions + (j - len(occupations)/2 + 0.5) * offset_width

        # Draw vertical line (stick)
        axes[i].vlines(x=pos, ymin=0, ymax=values,
                      color=colors[j],
                      alpha=1,
                      linewidth=8)

        # Draw the circle (head)
        axes[i].scatter(pos, values,
                        color='white',
                        s=900,
                        alpha=0.3,
                        edgecolors=colors[j],
                        zorder=2)
        axes[i].scatter(pos, values,
                        color=colors[j],
                        s=600,
                        edgecolors='white',
                        linewidth=1,
                        alpha=0.9,
                        label=occupation,
                        zorder=3)

```

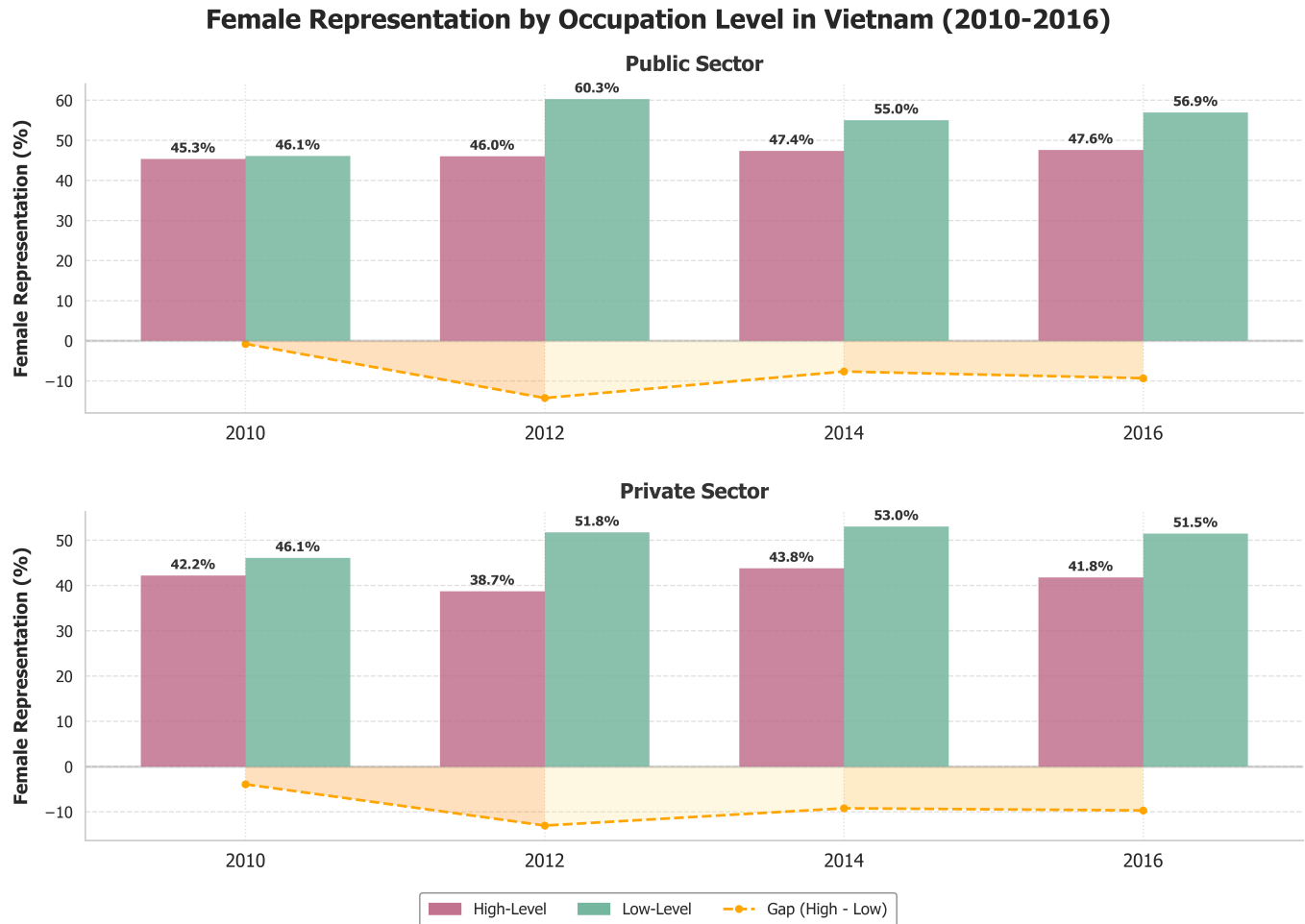


Figure 2: Female representation at high-skill and low-skill occupation levels in the public and private sectors in Vietnam from 2010 to 2016. The shaded area illustrates the gap between female representation in high-skill and low-skill occupations.

```
# Add value label
for (x, y) in zip(pos, values):
    axes[i].text(x, y, f'{int(round(y))}',
                 ha='center', va='center',
                 color='white', fontsize=10, fontweight='bold', zorder=4)
```

3.4 Discussion

The grouped bar chart with the gap line illustrated in Figure 2 shows that throughout 2010–2016, women consistently held a higher share of low-level occupations compared to high-level occupations in both sectors. Although there is some modest improvement in the public sector (with the gap narrowing slightly after 2012), the overall gap persists across time. In the private sector, the gap remains larger and relatively unchanged, indicating stronger structural barriers in promoting women to leadership positions. Whereas, the lollipop plot in Figure 3 provides more detailed insights at the occupation type level. Across both sectors, manager roles consistently have the lowest female representation, hovering around 20–30%. Meanwhile, clerical support roles and elementary occupations show the highest female participation rates, often exceeding 50% in the private sector. This suggests that while women participate actively in the workforce, they are disproportionately concentrated in lower-skilled, lower-paid positions. Additionally, the private sector appears less balanced across roles compared to the public sector. In the public sector, the distribution of female representation across different occupations is slightly more even, possibly reflecting government efforts to promote gender equality in public employment through policies and quotas.

The data reveal that while Vietnam has achieved notable levels of female participation in the workforce, particularly in lower-skill roles, there remains a persistent and measurable gender gap in access to high-skill positions. The widening

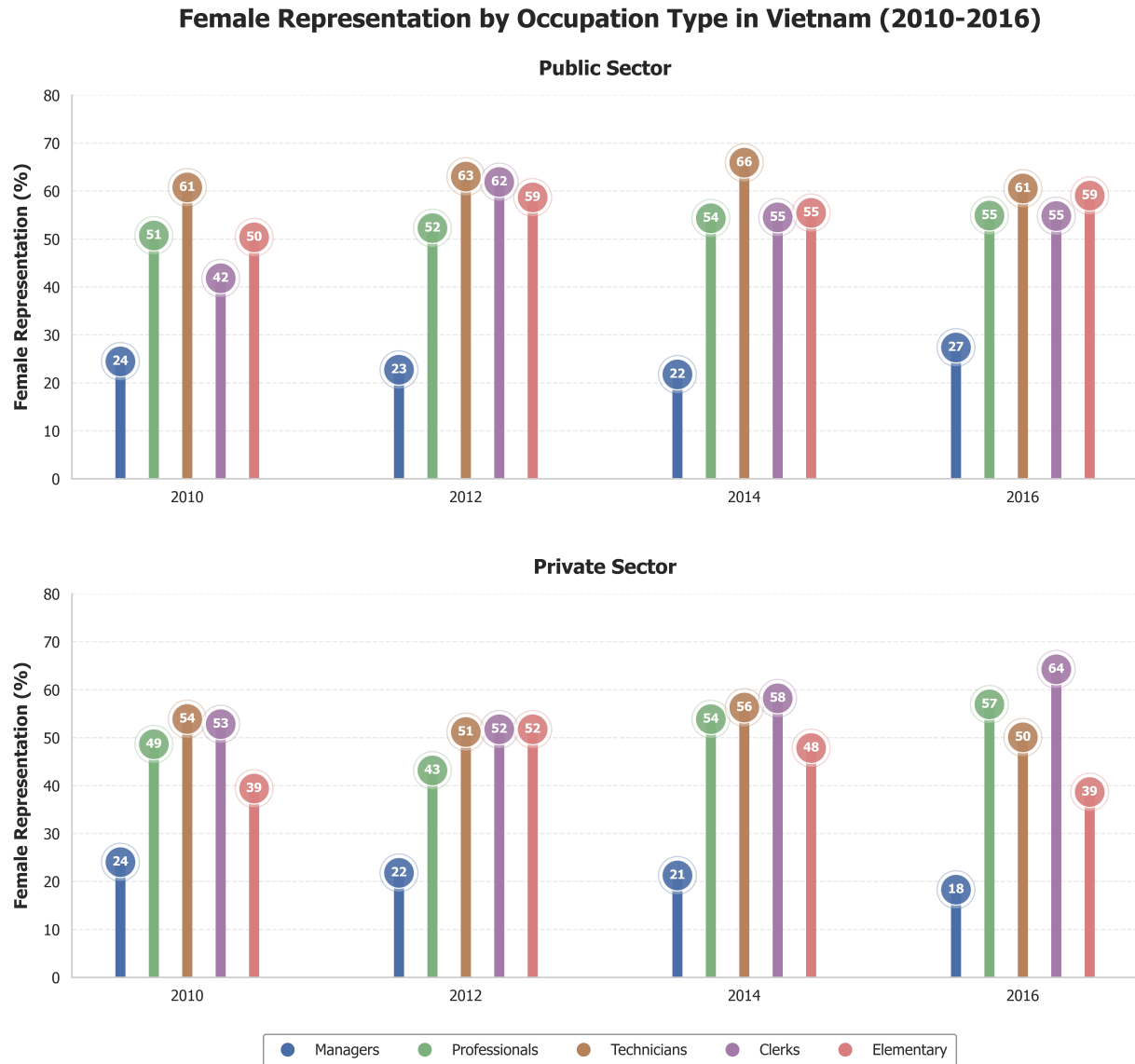


Figure 3: Female representation by detailed occupation types (managers, professionals, technicians, clerks, and elementary occupations) in the public and private sectors in Vietnam from 2010 to 2016.

of the gap in 2012 across both sectors may reflect broader socio-economic or policy-related disruptions, such as shifts in labor market demand or structural changes in public administration. These findings suggest structural barriers to female career advancement, including potential discrimination, limited mentorship, or unequal access to leadership training. Moreover, the concentration of women in clerical and elementary roles points to occupational segregation that may stem from societal norms or educational pathways. The visualizations offer not only a diagnostic tool but also a foundation for evidence-based policymaking aimed at fostering more equitable labor market outcomes in Vietnam.

4 Conclusion

In this project, we explored trends in public wage spending and gender equity in employment using the Worldwide Bureaucracy Indicators (WWBI) dataset. Our analysis revealed that public wage bills as a share of GDP vary considerably across regions and income groups, with high-income countries maintaining more stable levels and low-income countries showing greater volatility, often influenced by political and economic events. In Vietnam, we found persistent gender gaps in workforce participation, with women consistently underrepresented in higher-level occupations compared to lower-level ones, especially in the private sector. Although there have been modest improvements over time, the overall gap persists, indicating structural barriers to women's advancement in the workforce.

Given these findings, several additional steps could enhance both public sector management and workforce equity.

First, for public policy, governments — especially in developing economies, could benefit from benchmarking wage bill trends against peer countries to ensure sustainable public spending. In Vietnam, where public sector reform is a major topic of national debate, monitoring wage bills relative to GDP could help inform policies aimed at improving efficiency without jeopardizing service delivery. On the gender equity front, the analysis highlights the need for stronger policies promoting women’s advancement into high-level roles, particularly within the public sector. Policymakers could consider initiatives such as leadership training programs, transparent promotion processes, and supportive workplace policies (e.g., childcare support, flexible work arrangements) to foster a more inclusive environment.

While our project provides valuable insights, it is not without limitations. Data gaps and inconsistencies for some countries and time periods, particularly in earlier years, may affect the completeness and generalizability of our findings. Additionally, the Vietnam gender equity analysis was limited to only four survey years (2010–2016), which restricts our ability to capture longer-term trends. Future work could extend the analysis by incorporating more recent data if available, integrating contextual economic and political variables (e.g., GDP growth, education, gender equality indices) to better explain observed patterns, and expanding the gender equity study to more countries for regional comparison. Such extensions would enhance the robustness of the findings and support more informed public policy recommendations for sustainable governance and inclusive workforce development.

Overall, by linking fiscal management and gender equity challenges more closely to evidence from cross-national data, this project highlights pathways for developing more efficient, inclusive, and resilient public sectors.

References

- [1] *Turning promises into action: Gender equality in the 2030 agenda for sustainable development*. UN Women, 2018.
- [2] Muhammad Faisal Ali Baig, Zahid Hasnain, Daniel Oliver Rogger, and David Locke Newhouse. Worldwide bureaucracy indicators (wwbi): Codebook and explanatory note. 2023.
- [3] Data Science Learning Community. Tidy tuesday: A weekly social data project - worldwide bureaucracy indicators, 2024.